

Practice Problem

Imagine a child playing with a toy train. Each compartment of the train carries a single large **alphabet block** (like 'A', 'B', 'C').

The child constantly changes the train: adding blocks to the front, adding them to the back, inserting them in the middle, or removing blocks.

Your job is to write a program using a **Singly Linked List** to simulate this toy train. Each node in your list will represent a compartment and must store a char value.



Your Task

You need to create a struct for the nodes and write functions to handle the child's actions. Then, in your `main()` function, you must simulate the following sequence of events in order:

1. **Initialize:** Start with an empty train.
2. **Add to End:** The child connects block 'A', then 'C', then 'D' to the end of the train.
3. **Add to Front:** The child adds block 'Z' to the very front of the train.
4. **Insert After:** The child notices 'B' is missing. Insert block 'B' immediately **after** block 'A'.

5. **Update:** The child decides to swap the first block. Update block '**Z**' to become block '**E**'.
6. **Delete:** The child then removes the Block '**C**' off the train.
7. **Search:** The child wants to check if block '**B**' is in the train and then check if block '**C**' is in the train.
8. **Display (Print All):** You must print the current train formation **after every single step** so we can see the train changing over time.

Sample Output

```
--- Train Simulation Start ---
```

```
Added A, C, D to end:
```

```
Train: A -> C -> D -> NULL
```

```
Added Z to front:
```

```
Train: Z -> A -> C -> D -> NULL
```

```
Inserted B after A:
```

```
Train: Z -> A -> B -> C -> D -> NULL
```

```
Updated Z to E:
```

```
Train: E -> A -> B -> C -> D -> NULL
```

```
Delete C:
```

```
Train: E -> A -> B -> D -> NULL
```

```
Search: Found B? Yes | Found C? No
```

```
--- Final Train Formation ---
```

```
E -> A -> B -> D -> NULL
```